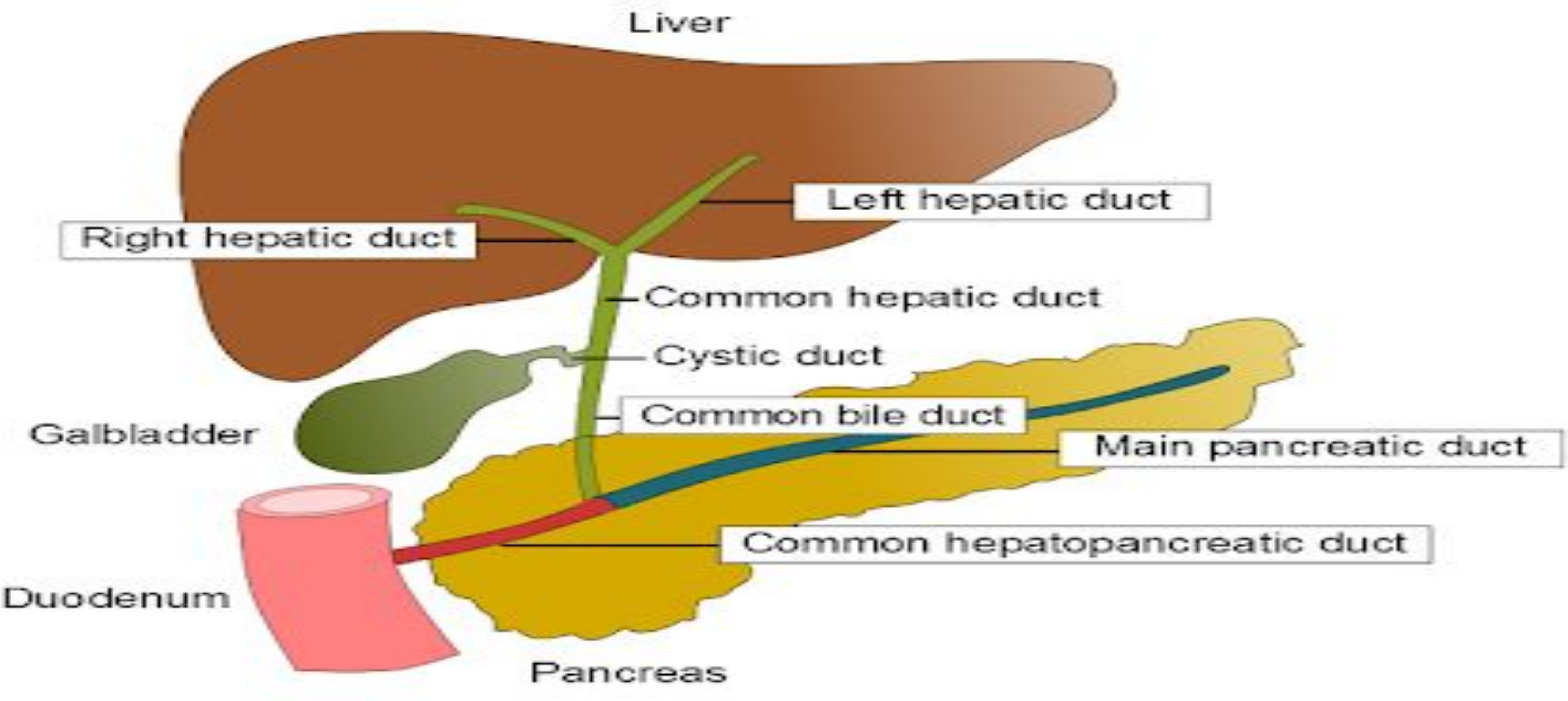
LIVER,BILE AND GALL BLADDER FUNCTIONS



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Learning Objectives

1-Functions of Liver.

2-Bile formation, secretion and functions.

3-Corelate bile to fat digestion.

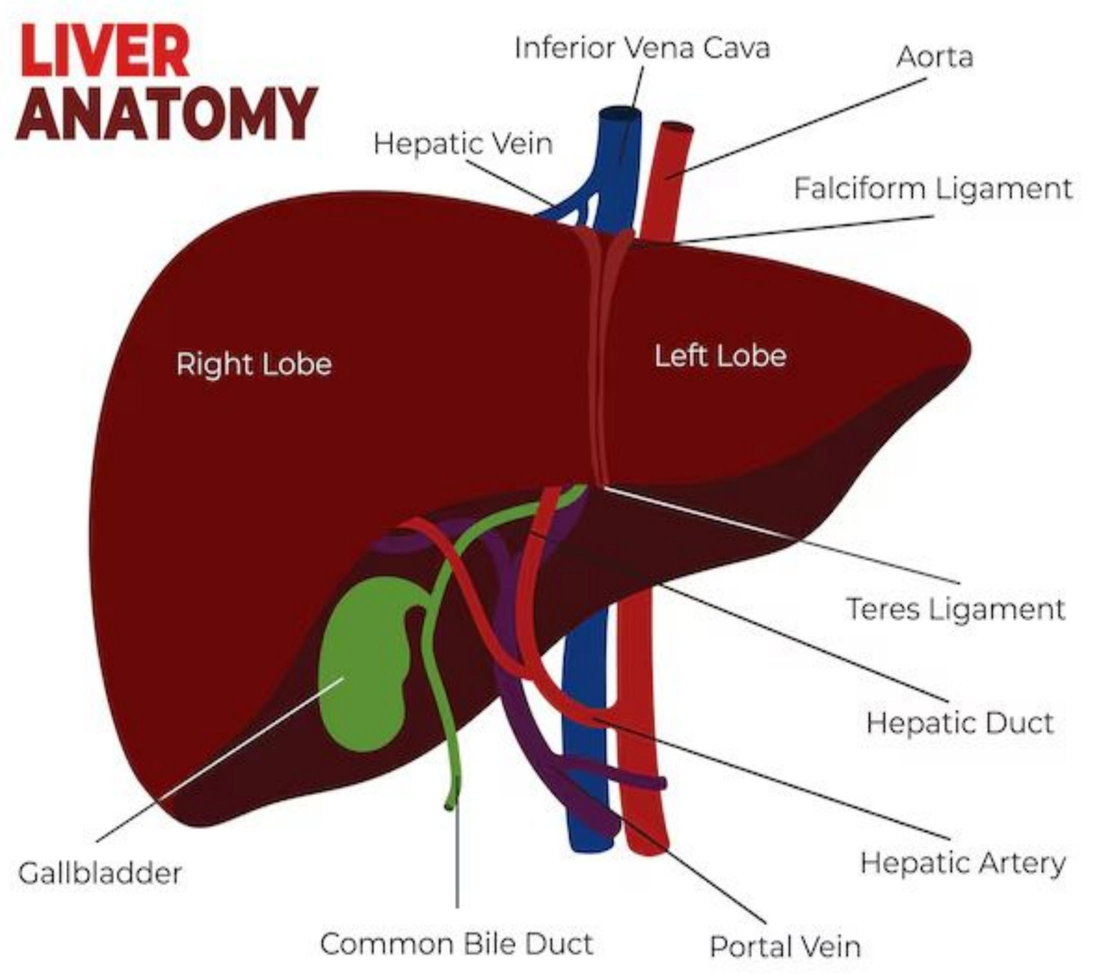
4-Gall bladder functions.

LIVER

The ***liver*** is the largest gland in the body.

It is located in the right, upper side of the abdominal cavity immediately beneath the diaphragm.

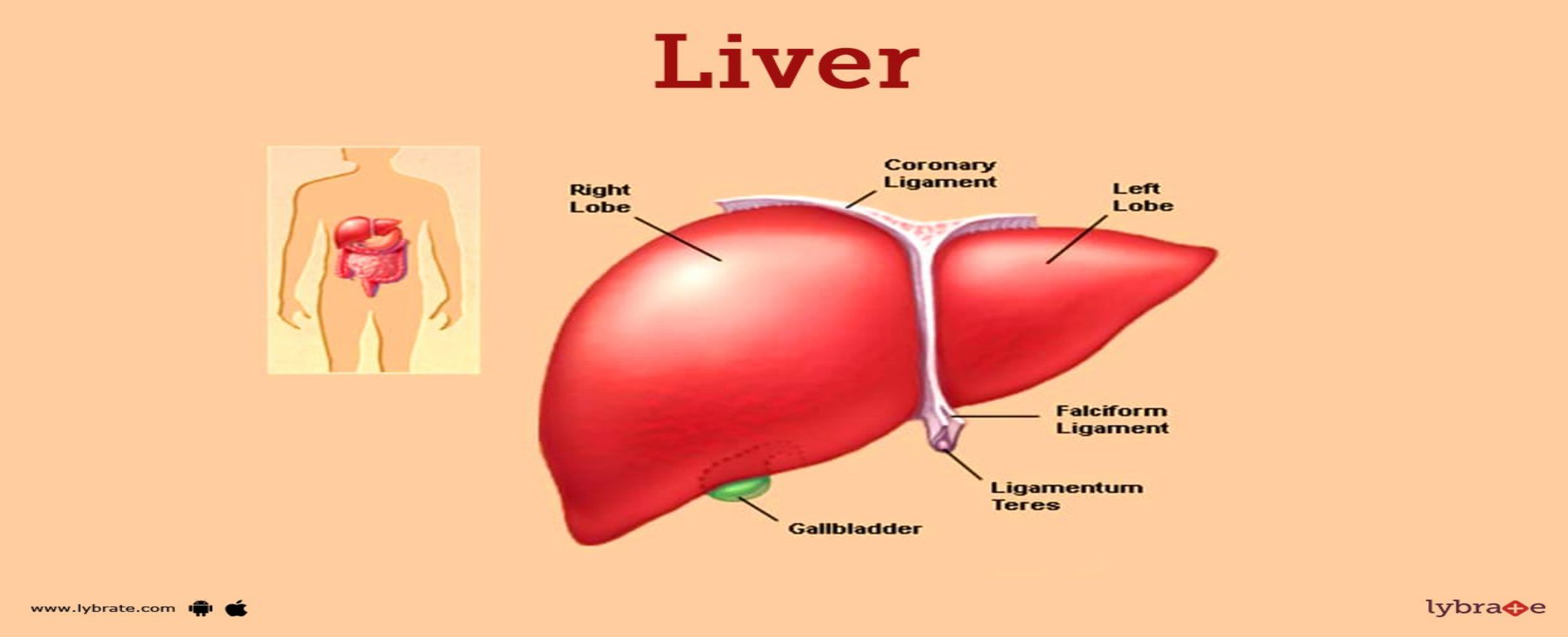
It weighs about three pounds and holds approximately 13 percent of the body’s blood supply.



Anatomy of the Liver

Consists of 2 lobes divided by falciform ligament

There is no known difference between the lobes



Liver

Basic structural unit is lobule, made of ramifying columns of hepatic cells.

Hepatic lobule is the structural and functional unit of liver.

There are about 50,000 to 100,000 lobules in the liver.

The lobule is a honeycomb-like structure and it is made up of liver cells called hepatocytes.

Hepatocytes

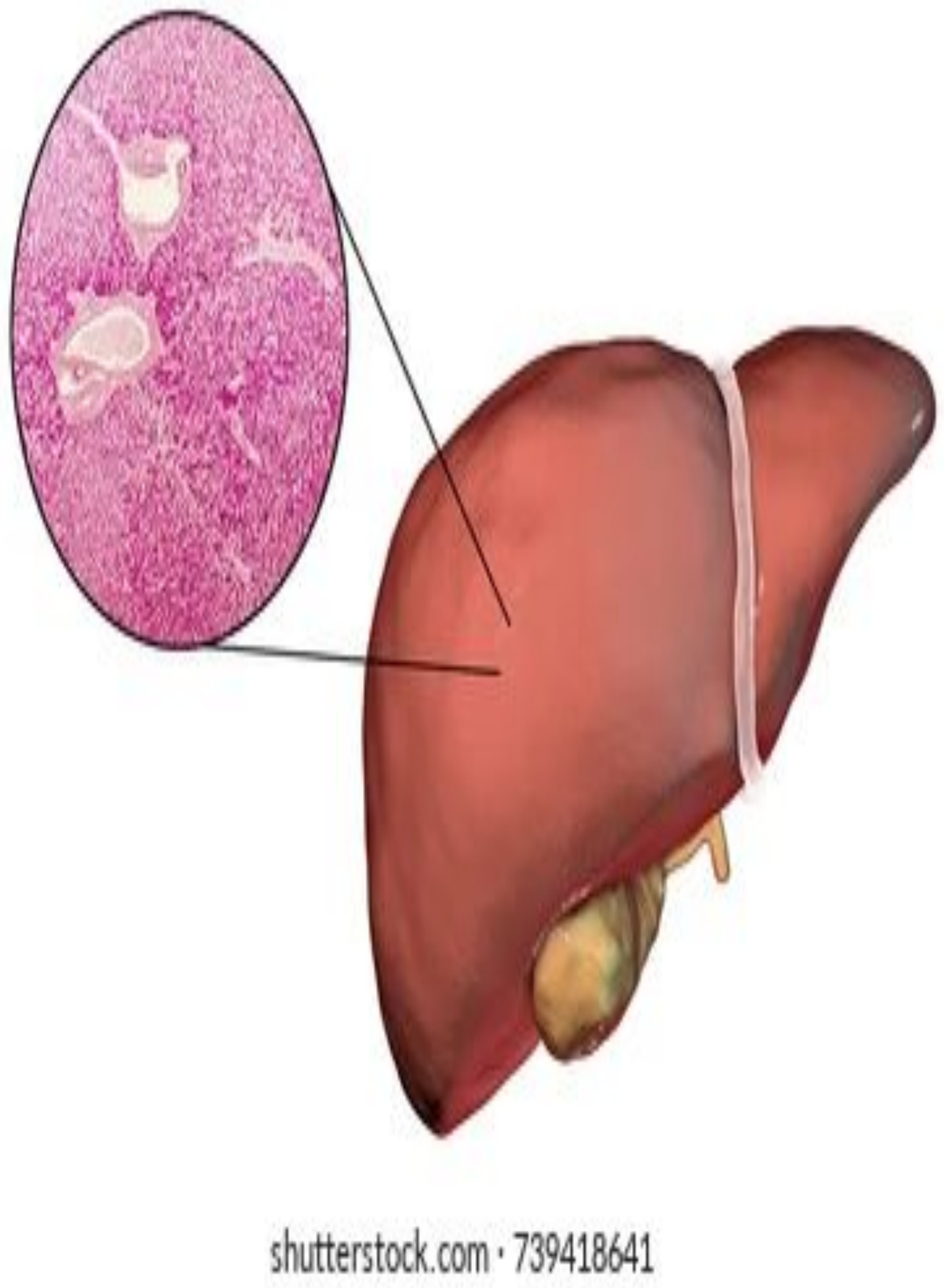
Hepatocyte are the primary cells in the liver that exchange substances with the blood filtering through the liver.

These hepatocytes are tunneled by a communicating system called – blood sinusoids which are lined by fenestrated endothelial cells.

The endothelial cells contain Kuepfer cells

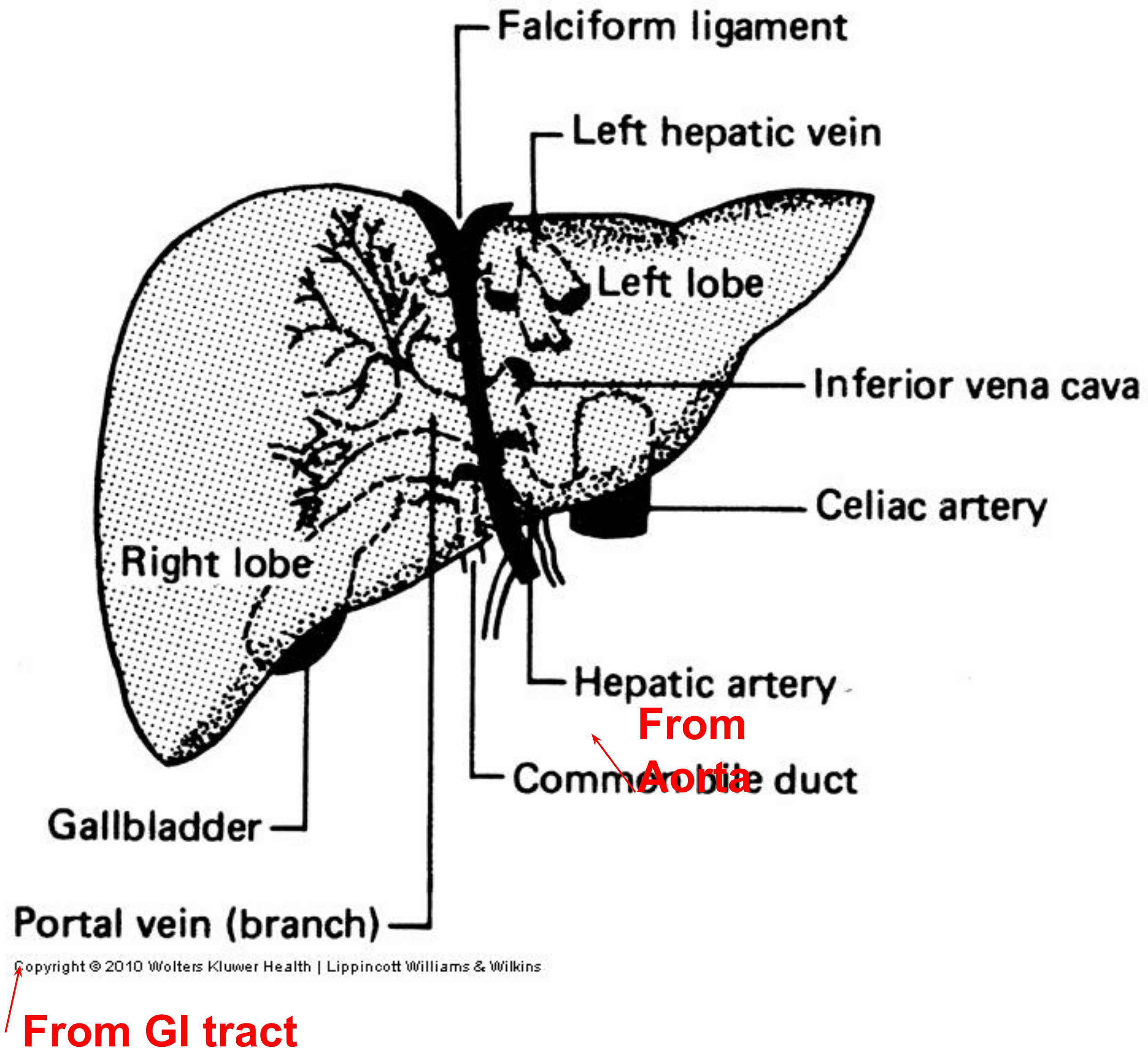
Portal vein and hepatic artery both empty into the sinusoids.

Sinusoids open into central vein.

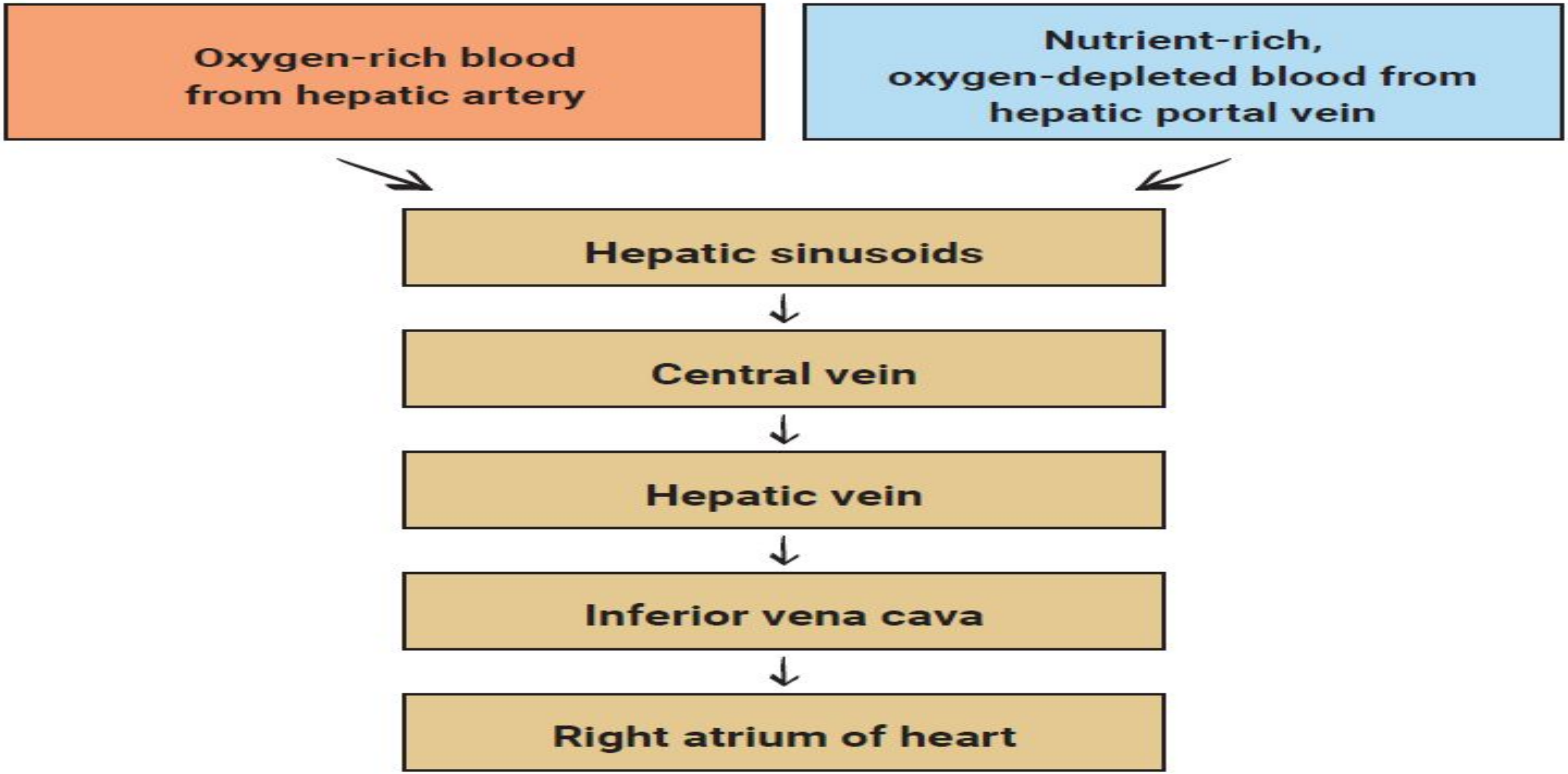


Liver

- Vascular organ
  - Hepatic artery
    - Supplies O_2 rich blood from heart to liver
    - Provides 20-30% of blood supply to liver
  - Portal vein
    - Supplies nutrient rich blood from the digestive tract
    - Provides 70-80% of blood to liver



Blood flow through liver



FUNCTIONS OF LIVER

The liver is the human body's largest internal organ and a vital metabolic hub.

It perform over 500 functions crucial for survival, including detoxification, digestion, and energy storage.

It filters over 250 gallons of blood daily, breaking down toxins, drugs, and waste products.

Key roles include producing bile for digestion, regulating blood sugar levels, and synthesizing essential proteins for blood clotting.

Liver is the largest gland and one of the vital organs of the body.

It performs many vital metabolic and homeostatic functions, which are summarized below.

1. METABOLIC FUNCTION

Liver is the organ where maximum metabolic reactions such as metabolism of carbohydrates, proteins, fats, vitamins and many hormones are carried out.

2.STORAGE FUNCTION

Many substances like glycogen, amino acids, iron, folic acid and vitamins A, B12 and D are stored in liver.

3.SYNTHETIC FUNCTION

Liver produces glucose by gluconeogenesis. It synthesizes all the plasma proteins and other proteins (except immunoglobulins) such as clotting factors, complement factors and hormone binding proteins. It also synthesizes steroids, somatomedin and heparin.

4.SECRETION OF BILE

Liver secretes bile which contains bile salts, bile pigments, cholesterol, fatty acids and lecithin.

The functions of bile are mainly due to bile salts.

Bile salts are required for digestion and absorption of fats in the intestine. Bile helps to carry away waste products and breakdown fats, which are excreted through feces or urine.

5.EXCRETORY FUNCTION

Liver excretes cholesterol, bile pigments, heavy metals (like lead, arsenic and bismuth), toxins, bacteria and virus (like that of yellow fever) through bile

6.HEAT PRODUCTION

Enormous amount of heat is produced in the liver because of metabolic reactions. Liver is the organ where maximum heat is produced.

7.HEMOPOIETIC FUNCTION

In fetus (hepatic stage), liver produces the blood cells.

It stores vitamin B12 necessary for erythropoiesis and iron necessary for synthesis of hemoglobin.

Liver produces thrombopoietin that promotes production of thrombocytes.

8.HEMOLYTIC FUNCTION

The senile RBCs after a lifespan of 120 days are destroyed by reticuloendothelial cells (Kupffer cells) of liver

9.INACTIVATION OF HORMONES AND DRUGS

Liver catabolizes the hormones such as growth hormone, parathormone, cortisol, insulin, glucagon and estrogen.

It also inactivates the drugs, particularly the fat-soluble drugs.

The fat-soluble drugs are converted into water soluble substances, which are excreted through bile or urine.

10-Detoxification & Filtration:

The liver filters all blood from the digestive tract, removing toxic substances, breaking down medications, and processing worn-out blood cells.

Detoxification includes drugs and poisons, and metabolic products like ammonia, alcohol, and bilirubin

11-Blood Sugar Management:

Regulates blood glucose levels by storing excess glucose as glycogen and releasing it when needed.

12-Immunity & Storage:

Acts as a key player in the immune system, removing bacteria from the bloodstream, and stores glucose, iron, and vitamins A, D, E, K, and B12.

PROPERTIES AND COMPOSITION OF BILE

PROPERTIES OF BILE

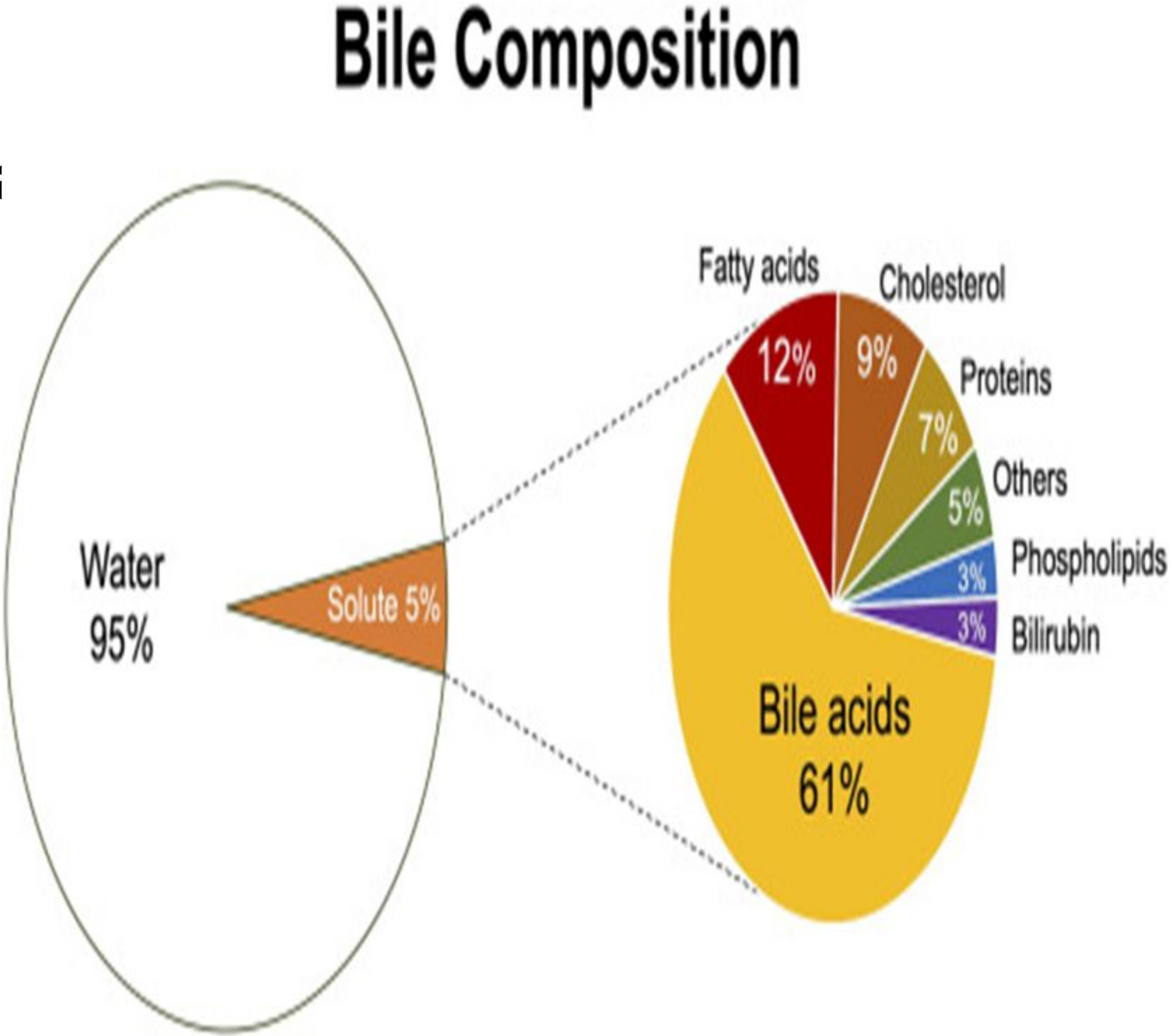
Volume : 800 to 1,200 mL/da

Reaction Alkaline

pH : 8 to 8.6

Specific gravity : 1.010 to 1.011

Color : Golden yellow or green.



SECRETION OF BILE

Bile is a yellow-green digestive fluid continuously produced by liver cells (hepatocytes) and modified by ductal cells, totaling about 800 to 1,000 milliliters daily.

Bile is secreted by hepatocytes. The initial bile secreted by hepatocytes contains large quantity of bile acids, bile pigments, cholesterol, lecithin and fatty acids.

Bile flows through the hepatic ducts to the gallbladder, where it is stored and concentrated by 5 to 10 times by removing water and electrolytes.

From hepatocytes, bile is released into canaliculi**,(Bile Canaliculi (Liver): These are narrow tubes formed by the apical membranes of neighboring hepatocytes. They collect bile produced by liver cells and transport it toward larger bile ducts)**.

From here, it passes through small ducts and hepatic ducts and reaches the common hepatic duct.

From common hepatic duct, bile is diverted either directly into the intestine or into the gallbladder.

STORAGE OF BILE

Most of the bile from liver enters the gallbladder, where it is stored.

It is released from gallbladder into the intestine whenever it is required.

When bile is stored in gallbladder it undergoes many changes both in quality and quantity such as:

1. Volume is decreased because of absorption of a large amount of water and electrolytes (except calcium and potassium)
2. Concentration of bile salts, bile pigments, cholesterol, fatty acids and lecithin is increased because of absorption of water and electrolytes
3. The pH is decreased slightly
4. Specific gravity is increased
5. Mucin is added to bile

BILE SALTS

Bile salts are essential digestive compounds produced by the liver from cholesterol and stored in the gallbladder to aid in the digestion and absorption of dietary fats and fat-soluble vitamins (A, D, E, K).

They act as emulsifiers, breaking down large fat globules into smaller particles (micelles) in the small intestine.

FORMATION OF BILE SALTS

Bile salts are formed from bile acids. There are two primary bile acids in human, namely **cholic acid and chenodeoxycholic acid,** which are formed in liver and enter the intestine through bile.

Due to the bacterial action in the intestine, the primary bile acids are converted into secondary bile acids:

- Cholic acid ⟶ deoxycholic acid
- Chenodeoxycholic acid ⟶ lithocholic acid

Secondary bile acids from intestine are transported back to liver through enterohepatic circulation.

FUNCTIONS OF BILE SALTS

Bile salts are required for digestion and absorption of fats in the intestine. The functions of bile salts are:

1.Emulsification of Fats

Emulsification is the process by which the fat globules are broken down into minute droplets and made in the form of a milky fluid called emulsion in small intestine, by the action of bile salts.

Bile salts emulsify the fats by reducing the surface tension due to their detergent action. Now the fats can be easily digested by lipolytic enzymes

Unemulsified fat usually passes through the intestine and then it is eliminated in feces. Emulsification of fats by bile salts needs the presence of lecithin from bile.

2-Absorption of Fats

Bile salts help in the absorption of digested fats from intestine into blood. Bile salts combine with fats and make complexes of fats called micelles. The fats in the form of micelles can be absorbed easily.

3. Cholagogue Action

Cholagogue is an agent which causes contraction of gallbladder and release of bile into the intestine.

Bile salts act as cholagogues indirectly by stimulating the secretion of hormone cholecystokinin.

This hormone causes contraction of gallbladder, resulting in release of bile.

5. Laxative Action

Laxative is an agent which induces defecation. Bile salts act as laxatives by stimulating peristaltic movements of the intestine.

6.Prevention of Gallstone Formation

Bile salts prevent the formation of gallstone by keeping the cholesterol and lecithin in solution.

In the absence of bile salts, cholesterol precipitates along with lecithin and forms gallstone

FUNCTIONS OF BILE

Most of the functions of bile are due to the bile salts.

1-DIGESTIVE FUNCTION

Lipolytic enzymes of GI tract cannot digest the fats directly because the fats are insoluble in water due to the surface tension

Bile salts break large fat globules into tiny droplets (emulsification).This greatly increases the surface area for lipase enzymes, produced in the pancreas, to digest the fats efficiently.

2.Absorption of Fats and Vitamins:

Bile salts help in the absorption of digested fats from intestine into blood. Bile salts combine with fats and make complexes of fats called micelles. The fats in the form of micelles can be absorbed easily.

Bile aids in the absorption of digested fat-soluble substances, including fatty acids and vitamins A, D, E, and K, through the intestinal wall.

3. EXCRETORY FUNCTIONS

Bile pigments are the major excretory products of the bile. Other substances excreted in bile are:

i. Heavy metals like copper and iron

ii. Some bacteria like typhoid bacteria

iii. Some toxins

iv. Cholesterol

v. Lecithin

vi. Alkaline phosphatase.

4. LAXATIVE ACTION

Bile salts act as laxatives Bile salts act as laxatives by stimulating peristaltic movements of the intestine. Increased bile flow acts as a natural laxative to aid digestion and bowel elimination.

5. ANTISEPTIC ACTION

Bile inhibits the growth of certain bacteria in the lumen of intestine by its natural detergent action.

6.CHOLERETIC ACTION

Choleretics are substances, including herbs and medications, that increase bile production in the liver, stimulating bile flow and enhancing fat digestion

7.MAINTENANCE OF pH IN GASTROINTESTINAL TRACT

As bile is highly alkaline, it neutralizes the acid chyme which enters the intestine from stomach. Thus, an optimum pH is maintained for the action of digestive enzymes.

8-Prevention of Gallstone formation

In the absence of bile salts, cholesterol precipitates along with lecithin and forms gallstone

9-Excretion of Waste Products:

The liver uses bile to eliminate waste products, specifically bilirubin (a breakdown product of red blood cells) and excess cholesterol, which are then excreted in feces.

10-Neutralization of Stomach Acid:

Bile is alkaline (pH 7.50 to 8.05), which allows it to neutralize the acidic chyme coming from the stomach, creating an optimal pH for pancreatic enzymes to function in the small intestine.

11.LUBRICATION FUNCTION

The mucin in bile acts as a lubricant for the chyme in intestine.

12.CHOLAGOGUE ACTION

A cholagogue is an agent—often herbal—that stimulates gallbladder contraction to release stored bile into the duodenum (small intestine).

This action enhances digestion (specifically fats), supports liver detoxification, and aids in the elimination of waste products like excess cholesterol.

GALLBLADDER

The gallbladder is a small, pear-shaped organ located in the upper right abdomen beneath the liver that stores and concentrates bile, a fluid produced by the liver to aid in digesting fats.

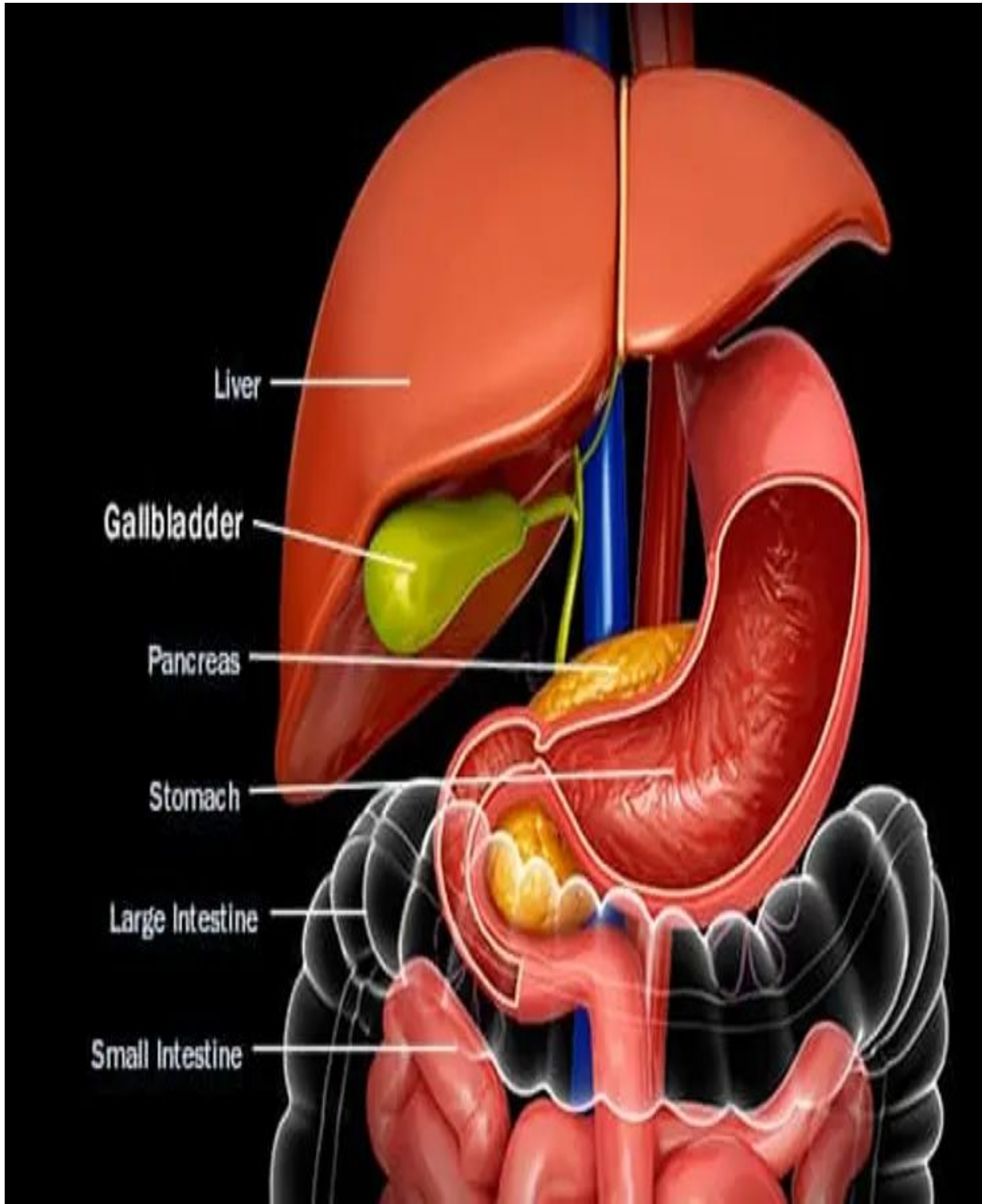
Bile secreted from liver is stored in gallbladder.

The capacity of gallbladder is approximately 50 mL.

Gallbladder is not essential for life and it is removed (cholecystectomy) in patients suffering from gallbladder dysfunction.

After cholecystectomy, patients do not suffer from any major disadvantage.

In some species, gallbladder is absent



FUNCTIONS OF GALLBLADDER

Major functions of gallbladder are the storage and concentration of bile.

1.Storage of Bile

Bile is continuously secreted from liver. But it is released into intestine only intermittently and most of the bile is stored in gallbladder till it is required.

2.Concentration of Bile

Bile is concentrated while it is stored in gallbladder.

The gallbladder concentrates bile by up to 10-20 times by absorbing water and electrolytes, making it more effective for digestion.

The mucosa of gallbladder rapidly reabsorbs water and electrolytes, except calcium and potassium. But the bile salts, bile pigments, cholesterol and lecithin are not reabsorbed.

3.Alteration of pH of Bile

The pH of bile decreases from 8 – 8.6 to 7 – 7.6 and it becomes less alkaline when it is stored in gallbladder.

4.Secretion of Mucin

Gallbladder secretes mucin and adds it to bile. When bile is released into the intestine, mucin acts as a lubricant for movement of chyme in the intestine.

5.Maintenance of Pressure in Biliary System

Due to the concentrating capacity, gallbladder maintains a pressure of about 7 cm H_2O in biliary system. This pressure in the biliary system is essential for the release of bile into the intestine.

Digestion Aid:

It releases bile into the duodenum (small intestine) in response to food consumption, specifically for digesting fats.

Controlled Release:

It serves as a reservoir that releases a concentrated "bolus" of bile when needed, rather than the slow, continuous flow from the liver.

FILLING AND EMPTYING OF GALLBLADDER

The gallbladder stores and concentrates bile produced by the liver, filling during fasting and contracting to empty into the duodenum after meals.

This process is driven by the hormone CCK and regulates digestion.

1-Filling of gallbladder

It holds 40 to 50 ml of bile, concentrating it 5 to 10 times by removing water and electrolytes.

The gallbladder fills with bile produced by the liver, which is diverted to it when the sphincter of Oddi (a valve near the small intestine) is closed, such as between meals.

It holds up to 50 ml of fluid, but concentrates bile 5–10 times, allowing it to store the bile secreted over several hours.

2-Gallbladder Emptying (Fed State)

Gallbladder emptying is the process where the gallbladder contracts, stimulated primarily by the hormone cholecystokinin (CCK) after eating, to release stored bile into the duodenum for fat digestion.

This process involves contraction of the gallbladder wall and relaxation of the sphincter of Oddi, often triggered by fatty foods.

REGULATION OF BILE SECRETION

Bile secretion is a continuous process though the amount is less during fasting. It starts increasing after meals and continues for three hours. Secretion of bile from liver and release of bile from the gallbladder are influenced by some chemical factors, which are categorized into three groups

1. Choleretics
2. Cholagogue
3. Hydrocholeretic agents.

1. Choleretics

Substances which increase the secretion of bile from liver are known as choleretics.

Effective choleretic agents are:

i. Acetylcholine

ii. Secretin

iii. Cholecystokinin

iv. Acid chyme in intestine

v. Bile salts.

2.Cholagogues

Cholagogue is an agent which increases the release of bile into the intestine by contracting gallbladder.

Common cholagogues are:

i. Bile salts

ii. Calcium

iii. Fatty acids

iv. Amino acids

v. Inorganic acids

All these substances stimulate the secretion of cholecystokinin, which in turn causes contraction of gallbladder and flow of bile into intestine.

3. Hydrocholeretic Agents

Hydrocholeretic agent is a substance which causes the secretion of bile from liver, with large amount of water and less amount of solids. Hydrochloric acid is a hydrocholeretic agent.